

Problem 1: Online Shopping System

Scenario

An e-commerce company wants to maintain customer information and their product orders.

Students need to:

- Create the tables.
- Insert the following records manually.
- Practice UPDATE, DELETE and JOIN operations.

Table 1: Customers

Customer ID	Customer Name	City	Phone
101	Amit Sharma	Delhi	9876543201
102	Priya Verma	Mumbai	9876543202
103	Rahul Singh	Pune	9876543203
104	Sneha Gupta	Jaipur	9876543204
105	Arjun Patel	Ahmedabad	9876543205
106	Neha Joshi	Surat	9876543206
107	Karan Mehta	Chandigarh	9876543207
108	Pooja Nair	Kochi	9876543208
109	Rohan Das	Kolkata	9876543209
110	Anjali Roy	Lucknow	9876543210
111	Vivek Kumar	Patna	9876543211
112	Meera Shah	Indore	9876543212
113	Deepak Yadav	Nagpur	9876543213
114	Nisha Jain	Bhopal	9876543214
115	Mohit Saini	Noida	9876543215
116	Aisha Khan	Hyderabad	9876543216
117	Saurabh Mishra	Kanpur	9876543217
118	Ritu Agarwal	Agra	9876543218
119	Akash Malhotra	Ludhiana	9876543219
120	Simran Kaur	Amritsar	9876543220
121	Harsh Vyas	Udaipur	9876543221
122	Divya Rao	Mysore	9876543222
123	Manish Saxena	Dehradun	9876543223
124	Kavita Singh	Ranchi	9876543224
125	Aditya Bose	Bhubaneswar	9876543225

Table 2: Orders

Order ID	Customer ID	Product Name	Quantity	Price (₹)	Order Date
1001	101	Laptop	1	65000	2026-01-10
1002	102	Smartphone	2	25000	2026-01-11
1003	103	Headphones	3	2500	2026-01-12
1004	104	Keyboard	2	1800	2026-01-12
1005	105	Mouse	4	700	2026-01-13
1006	106	Monitor	1	15000	2026-01-14
1007	107	Printer	1	12000	2026-01-15
1008	108	Tablet	2	28000	2026-01-16
1009	109	Webcam	3	2200	2026-01-17

1010	110	SSD	2	5000	2026-01-18
1011	111	Laptop	1	72000	2026-01-19
1012	112	Smartphone	1	30000	2026-01-20
1013	113	Smart Watch	2	8500	2026-01-21
1014	114	Bluetooth Speaker	2	3500	2026-01-22
1015	115	Router	1	2800	2026-01-23
1016	116	Power Bank	3	1800	2026-01-24
1017	117	Pendrive	5	900	2026-01-25
1018	118	External HDD	2	6500	2026-01-26
1019	119	Gaming Mouse	1	3200	2026-01-27
1020	120	Mechanical Keyboard	1	5500	2026-01-28
1021	121	Laptop Bag	2	1800	2026-01-29
1022	122	UPS	1	6500	2026-01-30
1023	123	Graphics Card	1	42000	2026-01-31
1024	124	CPU Cabinet	1	4800	2026-02-01
1025	125	Projector	1	35000	2026-02-02

Practice Questions

CREATE

- a) Create a database named **OnlineShoppingDB**.

```
mysql> create database Online_shopping_system;
Query OK, 1 row affected (0.12 sec)
```

- b) Create the **Customers** table.

```
mysql> create table Customer(costumeID int primary key,customerName varchar(50) not
null,city varchar(30),phone varchar(15));
mysql> desc Customer;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| costumeID  | int           | NO   | PRI | NULL    |       |
| customerName | varchar(50)   | NO   |     | NULL    |       |
| city       | varchar(30)   | YES  |     | NULL    |       |
| phone      | varchar(15)   | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.03 sec)
```

- c) Create the **Orders** table with an appropriate foreign key.

```
mysql> create table Orders(orderID int primary key,costumeID int ,ProductName varchar
(40) not null, quantity int, price int,orderDate date,foreign key(costumeID) refe
rences Customer(costumeID));
mysql> desc Orders;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| orderID    | int           | NO   | PRI | NULL    |       |
| costumeID  | int           | YES  | MUL | NULL    |       |
| ProductName | varchar(40)   | NO   |     | NULL    |       |
| quantity   | int           | YES  |     | NULL    |       |
| price      | int           | YES  |     | NULL    |       |
| orderDate  | date          | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)
```

INSERT

- a) Insert all 25 customer records.

```
mysql> select *from Customer;
```

costumeID	customerName	city	phone
101	Amit Sharma	Delhi	9876543201
102	Priya Verma	Mumbai	9876543202
103	Rahul Singh	Pune	9876543203
104	Sneha Gupta	Jaipur	9876543204
105	Arjun Patel	Ahmedabad	9876543205
106	Neha Joshi	Surat	9876543206
107	Karan Mehta	Chandigarh	9876543207
108	Pooja Nair	Kochi	9876543208
109	Rohan Das	Kolkata	9876543209
110	Anjali Roy	Lucknow	9876543210
111	Vivek Kumar	Patna	9876543211
112	Meera Shah	Indore	9876543212
113	Deepak Yadav	Nagpur	9876543213
114	Nisha Jain	Bhopal	9876543214
115	Mohit Saini	Noida	9876543215
116	Aisha khan	Hyderabad	9876543216
117	Saurabh Mishra	Kanpur	9876543217
118	Ritu Agarwal	Agra	9876543218
119	Akash Malhotra	Ludhiana	9876543219
120	Simran Kaur	Amritsar	9876543220
121	Harsh Kaur	Udaipur	9876543221
122	Divya Rao	Mysore	9876543222
123	Manish Saxena	Dehradun	9876543223
124	Kavita Singh	Ranchi	9876543224
125	Aditya Bose	Bhubaneswar	9876543225

25 rows in set (0.00 sec)

- b) Insert all 25 order records.

```
mysql> select * from Orders;
```

orderID	costumeID	ProductName	quantity	price	orderDate
1001	101	Laptop	1	65000	2026-01-10
1002	102	Smartphone	2	25000	2026-01-11
1003	103	Headphones	3	2500	2026-01-12
1004	104	Keyboard	2	1800	2026-01-12
1005	105	Mouse	4	700	2026-01-13
1006	106	Monitor	1	15000	2026-01-14
1007	107	Printer	1	12000	2026-01-15
1008	108	Tablet	2	28000	2026-01-16
1009	109	Webcam	3	2200	2026-01-17
1010	110	SSD	2	5000	2026-01-18
1011	111	Laptop	1	72000	2026-01-19
1012	112	Smartphone	1	30000	2026-01-20
1013	113	Smart Watch	2	8500	2026-01-21
1014	114	Bluetooth Speaker	2	3500	2026-01-22
1015	115	Router	1	2800	2026-01-23
1016	116	PowerBank	3	1800	2026-01-24
1017	117	Pendrive	5	900	2026-01-25
1018	118	External HDD	2	6500	2026-01-26
1019	119	Gaming Mouse	1	3200	2026-01-27
1020	120	Mechanical Keyboard	1	5500	2026-01-28
1021	121	Laptop Bag	2	1800	2026-01-29
1022	122	UPS	1	6500	2026-01-30
1023	123	Graphics Card	1	42000	2026-01-31
1024	124	CPU Cabinet	1	4800	2026-01-01
1025	125	Projector	1	35000	2026-01-02

25 rows in set (0.00 sec)

UPDATE

- Change the city of **Customer ID 115** from **Noida** to **Gurugram**.
- Update the phone number of **Customer ID 122**.
- Change the quantity of **Order ID 1017** from **5** to **8**.
- Increase the price of the **Projector** to **₹37,000**.
- Change the product name **Router** to **WiFi Router**.

```
mysql> update Customer set city = 'Gurugram' where costumeID = 115;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> update Customer set phone = '9876543453' where costumeID = 122;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> update Orders set quantity = 8 where orderID = 1017;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> update Orders set price = 37000 where orderID = 1025;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> update Orders set ProductName = 'Wifi Router' where orderID = 1015;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'Router where
orderID = 1015' at line 1
mysql> update Orders set ProductName = 'Wifi Router' where orderID = 1015;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> select * from Customer;
```

costumeID	customerName	city	phone
101	Amit Sharma	Delhi	9876543201
102	Priya Verma	Mumbai	9876543202
103	Rahul Singh	Pune	9876543203
104	Sneha Gupta	Jaipur	9876543204
105	Arjun Patel	Ahmedabad	9876543205
106	Neha Joshi	Surat	9876543206
107	Karan Mehta	Chandigarh	9876543207
108	Pooja Nair	Kochi	9876543208
109	Rohan Das	Kolkata	9876543209
110	Anjali Roy	Lucknow	9876543210
111	Vivek Kumar	Patna	9876543211
112	Meera Shah	Indore	9876543212
113	Deepak Yadav	Nagpur	9876543213
114	Nisha Jain	Bhopal	9876543214
115	Mohit Saini	Gurugram	9876543215
116	Aisha Khan	Hyderabad	9876543216
117	Saurabh Mishra	Kanpur	9876543217
118	Ritu Agarwal	Agra	9876543218
119	Akash Malhotra	Ludhiana	9876543219
120	Simran Kaur	Amritsar	9876543220
121	Harsh Kaur	Udaipur	9876543221
122	Divya Rao	Mysore	9876543453
123	Manish Saxena	Dehradun	9876543223
124	Kavita Singh	Ranchi	9876543224
125	Aditya Bose	Bhubaneswar	9876543225

```
25 rows in set (0.00 sec)
mysql> select * from Orders;
```

orderID	costumeID	ProductName	quantity	price	orderDate
1001	101	Laptop	1	65000	2026-01-10
1002	102	Smartphone	2	25000	2026-01-11
1003	103	Headphones	3	2500	2026-01-12
1004	104	Keyboard	2	1800	2026-01-12
1005	105	Mouse	4	700	2026-01-13
1006	106	Monitor	1	15000	2026-01-14
1007	107	Printer	1	12000	2026-01-15
1008	108	Tablet	2	28000	2026-01-16
1009	109	Webcam	3	2200	2026-01-17
1010	110	SSD	2	5000	2026-01-18
1011	111	Laptop	1	72000	2026-01-19
1012	112	Smartphone	1	30000	2026-01-20
1013	113	Smart Watch	2	8500	2026-01-21
1014	114	delete from OrderID = 125;	2	3500	2026-01-22
1015	115	Wifi Router	1	2800	2026-01-23
1016	116	PowerBank	3	1800	2026-01-24
1017	117	Pendrive	8	900	2026-01-25
1018	118	External HDD	2	6500	2026-01-26
1019	119	Gaming Mouse	1	3200	2026-01-27
1020	120	Mechanical Keyboard	1	5500	2026-01-28
1021	121	Laptop Bag	2	1800	2026-01-29
1022	122	UPS	1	6500	2026-01-30
1023	123	Graphics Card	1	42000	2026-01-31
1024	124	CPU Cabinet	1	4800	2026-01-01
1025	125	Projector	1	37000	2026-01-02

```
25 rows in set (0.00 sec)
```

DELETE

- a) Delete **Customer ID 125**.
- b) Delete **Order ID 1025**.
- c) Delete the order for **Gaming Mouse**.
- d) Delete the customer whose name is **Harsh Vyas**.
- e) Delete the order placed on **2026-01-14**.

```
mysql> select * from Customer;
```

costumeID	customerName	city	phone
101	Amit Sharma	Delhi	9876543201
102	Priya Verma	Mumbai	9876543202
103	Rahul Singh	Pune	9876543203
104	Sneha Gupta	Jaipur	9876543204
105	Arjun Patel	Ahmedabad	9876543205
106	Neha Joshi	Surat	9876543206
107	Karan Mehta	Chandigarh	9876543207
108	Pooja Nair	Kochi	9876543208
109	Rohan Das	Kolkata	9876543209
110	Anjali Roy	Lucknow	9876543210
111	Vivek Kumar	Patna	9876543211
112	Meera Shah	Indore	9876543212
113	Deepak Yadav	Nagpur	9876543213
114	Nisha Jain	Bhopal	9876543214
115	Mohit Saini	Gurugram	9876543215
116	Aisha Khan	Hyderabad	9876543216
117	Saurabh Mishra	Kanpur	9876543217
118	Ritu Agarwal	Agra	9876543218
119	Akash Malhotra	Ludhiana	9876543219
120	Simran Kaur	Amritsar	9876543220
121	Harsh Kaur	Udaipur	9876543221
122	Divya Rao	Mysore	9876543453
123	Manish Saxena	Dehradun	9876543223
124	Kavita Singh	Ranchi	9876543224
125	Aditya Bose	Bhubaneswar	9876543225

25 rows in set (0.00 sec)

```
mysql> select * from Orders;
```

orderID	costumeID	ProductName	quantity	price	orderDate
1001	101	Laptop	1	65000	2026-01-10
1002	102	Smartphone	2	25000	2026-01-11
1003	103	Headphones	3	2500	2026-01-12
1004	104	Keyboard	2	1800	2026-01-12
1005	105	Mouse	4	700	2026-01-13
1007	107	Printer	1	12000	2026-01-15
1008	108	Tablet	2	28000	2026-01-16
1009	109	Webcam	3	2200	2026-01-17
1010	110	SSD	2	5000	2026-01-18
1011	111	Laptop	1	72000	2026-01-19
1012	112	Smartphone	1	30000	2026-01-20
1013	113	Smart Watch	2	8500	2026-01-21
1014	114	Bluetooth Speaker	2	3500	2026-01-22
1015	115	Wifi Router	1	2800	2026-01-23
1016	116	PowerBank	3	1800	2026-01-24
1017	117	Pendrive	8	900	2026-01-25
1018	118	External HDD	2	6500	2026-01-26
1020	120	Mechanical Keyboard	1	5500	2026-01-28
1022	122	UPS	1	6500	2026-01-30
1023	123	Graphics Card	1	42000	2026-01-31
1024	124	CPU Cabinet	1	4800	2026-01-01

JOIN

- a) Display customer name along with the product they ordered.

```
mysql> select Customer.customerName,Orders.ProductName from Customer join Orders on Customer.costumeID = Orders.costumeID;
```

customerName	ProductName
Amit Sharma	Laptop
Priya Verma	Smartphone
Rahul Singh	Headphones
Sneha Gupta	Keyboard
Arjun Patel	Mouse
Karan Mehta	Printer
Pooja Nair	Tablet
Rohan Das	Webcam
Anjali Roy	SSD
Vivek Kumar	Laptop
Meera Shah	Smartphone
Deepak Yadav	Smart Watch
Nisha Jain	Bluetooth Speaker
Mohit Saini	Wifi Router
Aisha Khan	PowerBank
Saurabh Mishra	Pendrive
Ritu Agarwal	External HDD
Simran Kaur	Mechanical Keyboard
Divya Rao	UPS
Manish Saxena	Graphics Card
Kavita Singh	CPU Cabinet

21 rows in set (0.00 sec)

- b) Display customer city along with product name.

```
mysql> select Customer.city,Orders.Productname from Customer join Orders on Customer.costumeID = Orders.costumeID;
```

city	Productname
Delhi	Laptop
Mumbai	Smartphone
Pune	Headphones
Jaipur	Keyboard
Ahmedabad	Mouse
Chandigarh	Printer
Kochi	Tablet
Kolkata	Webcam
Lucknow	SSD
Patna	Laptop
Indore	Smartphone
Nagpur	Smart Watch
Bhopal	Bluetooth Speaker
Gurugram	Wifi Router
Hyderabad	PowerBank
Kanpur	Pendrive
Agra	External HDD
Amritsar	Mechanical Keyboard
Mysore	UPS
Dehradun	Graphics Card
Ranchi	CPU Cabinet

21 rows in set (0.00 sec)

- c) Display order ID, customer name, product name, and order date.

```
mysql> select Orders.orderID,Customer.customerName,Orders.Productname,Orders.orderDate from Customer join Orders on Customer.costumeID = Orders.costumeID;
```

orderID	customerName	Productname	orderDate
1001	Amit Sharma	Laptop	2026-01-10
1002	Priya Verma	Smartphone	2026-01-11
1003	Rahul Singh	Headphones	2026-01-12
1004	Sneha Gupta	Keyboard	2026-01-12
1005	Arjun Patel	Mouse	2026-01-13
1007	Karan Mehta	Printer	2026-01-15
1008	Pooja Nair	Tablet	2026-01-16
1009	Rohan Das	Webcam	2026-01-17
1010	Anjali Roy	SSD	2026-01-18
1011	Vivek Kumar	Laptop	2026-01-19
1012	Meera Shah	Smartphone	2026-01-20
1013	Deepak Yadav	Smart Watch	2026-01-21
1014	Nisha Jain	Bluetooth Speaker	2026-01-22
1015	Mohit Saini	Wifi Router	2026-01-23
1016	Aisha Khan	PowerBank	2026-01-24
1017	Saurabh Mishra	Pendrive	2026-01-25
1018	Ritu Agarwal	External HDD	2026-01-26
1020	Simran Kaur	Mechanical Keyboard	2026-01-28
1022	Divya Rao	UPS	2026-01-30
1023	Manish Saxena	Graphics Card	2026-01-31
1024	Kavita Singh	CPU Cabinet	2026-01-01

21 rows in set (0.00 sec)

d) Display all orders along with customer phone numbers.

```
mysql> select Orders.orderID,Orders.ProductName,Orders.quantity,Orders.price,Orders.orderDate,Customer.phone from Customer inner join Orders on Customer.costumeID = Orders.costumeID;
```

orderID	ProductName	quantity	price	orderDate	phone
1001	Laptop	1	65000	2026-01-10	9876543201
1002	Smartphone	2	25000	2026-01-11	9876543202
1003	Headphones	3	2500	2026-01-12	9876543203
1004	Keyboard	2	1800	2026-01-12	9876543204
1005	Mouse	4	700	2026-01-13	9876543205
1007	Printer	1	12000	2026-01-15	9876543207
1008	Tablet	2	28000	2026-01-16	9876543208
1009	Webcam	3	2200	2026-01-17	9876543209
1010	SSD	2	5000	2026-01-18	9876543210
1011	Laptop	1	72000	2026-01-19	9876543211
1012	Smartphone	1	30000	2026-01-20	9876543212
1013	Smart Watch	2	8500	2026-01-21	9876543213
1014	Bluetooth Speaker	2	3500	2026-01-22	9876543214
1015	Wifi Router	1	2800	2026-01-23	9876543215
1016	PowerBank	3	1800	2026-01-24	9876543216
1017	Pendrivel	8	900	2026-01-25	9876543217
1018	External HDD	2	6500	2026-01-26	9876543218
1020	Mechanical Keyboard	1	5500	2026-01-28	9876543220
1022	UPS	1	6500	2026-01-30	9876543453
1023	Graphics Card	1	42000	2026-01-31	9876543223
1024	CPU Cabinet	1	4800	2026-01-01	9876543224

21 rows in set (0.00 sec)

e) Display customers who have placed orders for **Laptop**.

```
mysql> select Customer.customerName from Customer inner join Orders on Customer.costumeID=Orders.costumeID where Orders.ProductName='Laptop';
```

customerName
Amit Sharma
Vivek Kumar

2 rows in set (0.00 sec)

Problem 2: Hospital Management System

Scenario

A multi-specialty hospital wants to maintain information about its doctors and patients. Each patient is assigned to one doctor. Students need to create the tables, insert the records, and practice **CREATE**, **INSERT**, **UPDATE**, **DELETE**, and **JOIN** operations.

Table 1: Doctors

Doctor ID	Doctor Name	Specialization	Experience (Years)
201	Dr. Amit Sharma	Cardiologist	15
202	Dr. Priya Verma	Neurologist	12
203	Dr. Rahul Singh	Orthopedic	10
204	Dr. Sneha Gupta	Pediatrician	8
205	Dr. Arjun Patel	Dermatologist	9
206	Dr. Neha Joshi	Gynecologist	14
207	Dr. Karan Mehta	ENT Specialist	11
208	Dr. Pooja Nair	Ophthalmologist	7
209	Dr. Rohan Das	Psychiatrist	13
210	Dr. Anjali Roy	General Physician	16
211	Dr. Vivek Kumar	Urologist	9
212	Dr. Meera Shah	Oncologist	17
213	Dr. Deepak Yadav	Nephrologist	12
214	Dr. Nisha Jain	Pulmonologist	10
215	Dr. Mohit Saini	Endocrinologist	8
216	Dr. Aisha Khan	Gastroenterologist	11
217	Dr. Saurabh Mishra	Rheumatologist	14
218	Dr. Ritu Agarwal	Surgeon	18
219	Dr. Akash Malhotra	Cardiologist	6
220	Dr. Simran Kaur	Pediatrician	5
221	Dr. Harsh Vyas	Dentist	9

222	Dr. Divya Rao	Neurologist	13
223	Dr. Manish Saxena	Orthopedic	7
224	Dr. Kavita Singh	General Physician	15
225	Dr. Aditya Bose	Dermatologist	8

Table 2: Patients

Patient ID	Patient Name	Doctor ID	Disease	Admission Date
301	Aarav Gupta	201	Heart Disease	2026-02-01
302	Siya Sharma	202	Migraine	2026-02-02
303	Rohan Verma	203	Fracture	2026-02-03
304	Ananya Singh	204	Viral Fever	2026-02-04
305	Kabir Patel	205	Skin Allergy	2026-02-05
306	Ishita Joshi	206	Pregnancy Checkup	2026-02-06
307	Vihaan Mehta	207	Ear Infection	2026-02-07
308	Myra Nair	208	Eye Infection	2026-02-08
309	Arjun Das	209	Anxiety	2026-02-09
310	Kiara Roy	210	Diabetes	2026-02-10
311	Aditya Kumar	211	Kidney Stone	2026-02-11
312	Aditi Shah	212	Cancer Screening	2026-02-12
313	Dev Yadav	213	Kidney Infection	2026-02-13
314	Riya Jain	214	Asthma	2026-02-14
315	Yash Saini	215	Thyroid	2026-02-15
316	Mehak Khan	216	Ulcer	2026-02-16
317	Aryan Mishra	217	Arthritis	2026-02-17
318	Tanvi Agarwal	218	Appendicitis	2026-02-18
319	Kunal Malhotra	219	Heart Checkup	2026-02-19
320	Pari Kaur	220	Cold & Cough	2026-02-20
321	Om Vyas	221	Tooth Decay	2026-02-21
322	Diya Rao	222	Epilepsy	2026-02-22
323	Rudra Saxena	223	Knee Pain	2026-02-23
324	Saanvi Singh	224	High Fever	2026-02-24
325	Krish Bose	225	Acne	2026-02-25

Practice Questions

CREATE

- a) Create a database named **HospitalDB**.

```
mysql> CREATE DATABASE HospitalDB;
Query OK, 1 row affected (0.02 sec)

mysql>
mysql> USE HospitalDB;
Database changed
```


- b) Create the **Doctors** table.
c) Create the **Patients** table with an appropriate foreign key.

```
mysql> CREATE TABLE Doctors (  
->     doctor_id INT PRIMARY KEY,  
->     doctor_name VARCHAR(50) NOT NULL,  
->     specialization VARCHAR(50) NOT NULL,  
->     experience INT  
-> );  
Query OK, 0 rows affected (0.04 sec)  
  
mysql> CREATE TABLE Patients (  
->     patient_id INT PRIMARY KEY,  
->     patient_name VARCHAR(50) NOT NULL,  
->     doctor_id INT,  
->     disease VARCHAR(100),  
->     admission_date DATE,  
->     FOREIGN KEY (doctor_id)  
->     REFERENCES Doctors(doctor_id)  
-> );  
Query OK, 0 rows affected (0.07 sec)  
  
mysql> desc Doctors;  
+-----+-----+-----+-----+-----+-----+  
| Field      | Type      | Null | Key | Default | Extra |  
+-----+-----+-----+-----+-----+-----+  
| doctor_id  | int       | NO   | PRI | NULL    |       |  
| doctor_name | varchar(50) | NO   |     | NULL    |       |  
| specialization | varchar(50) | NO   |     | NULL    |       |  
| experience  | int       | YES  |     | NULL    |       |  
+-----+-----+-----+-----+-----+-----+  
4 rows in set (0.00 sec)  
  
mysql> desc Patients;  
+-----+-----+-----+-----+-----+-----+  
| Field      | Type      | Null | Key | Default | Extra |  
+-----+-----+-----+-----+-----+-----+  
| patient_id  | int       | NO   | PRI | NULL    |       |  
| patient_name | varchar(50) | NO   |     | NULL    |       |  
| doctor_id   | int       | YES  | MUL | NULL    |       |  
| disease     | varchar(100) | YES  |     | NULL    |       |  
| admission_date | date      | YES  |     | NULL    |       |  
+-----+-----+-----+-----+-----+-----+  
5 rows in set (0.00 sec)
```

INSERT

- a) Insert all 25 doctor record.

```
mysql> select * from Doctors;
```

doctor_id	doctor_name	specialization	experience
201	Dr. Amit Sharma	Cardiologist	15
202	Dr. Priya Verma	Neurologist	12
203	Dr. Rahul Singh	Orthopedic	10
204	Dr. Sneha Gupta	Pediatrician	8
205	Dr. Arjun Patel	Dermatologist	9
206	Dr. Neha Joshi	Gynecologist	14
207	Dr. Karan Mehta	ENT Specialist	11
208	Dr. Pooja Nair	Ophthalmologist	7
209	Dr. Rohan Das	Psychiatrist	13
210	Dr. Anjali Roy	General Physician	16
211	Dr. Vivek Kumar	Urologist	9
212	Dr. Meera Shah	Oncologist	17
213	Dr. Deepak Yadav	Nephrologist	12
214	Dr. Nisha Jain	Pulmonologist	10
215	Dr. Mohit Saini	Endocrinologist	8
216	Dr. Aisha Khan	Gastroenterologist	11
217	Dr. Saurabh Mishra	Rheumatologist	14
218	Dr. Ritu Agarwal	Surgeon	18
219	Dr. Akash Malhotra	Cardiologist	6
220	Dr. Simran Kaur	Pediatrician	5
221	Dr. Harsh Vyas	Dentist	9
222	Dr. Divya Rao	Neurologist	13
223	Dr. Manish Saxena	Orthopedic	7
224	Dr. Kavita Singh	General Physician	15
225	Dr. Aditya Bose	Dermatologist	8

```
25 rows in set (0.00 sec)
```

- b) Insert all 25 patient records.

```
mysql> select * from Patients;
```

patient_id	patient_name	doctor_id	disease	admission_date
301	Aarav Gupta	201	Heart Disease	2026-02-01
302	Siya Sharma	202	Migraine	2026-02-02
303	Rohan Verma	203	Fracture	2026-02-03
304	Ananya Singh	204	Viral Fever	2026-02-04
305	Kabir Patel	205	Skin Allergy	2026-02-05
306	Ishita Joshi	206	Pregnancy Checkup	2026-02-06
307	Vihaan Mehta	207	Ear Infection	2026-02-07
308	Myra Nair	208	Eye Infection	2026-02-08
309	Arjun Das	209	Anxiety	2026-02-09
310	Kiara Roy	210	Diabetes	2026-02-10
311	Aditya Kumar	211	Kidney Stone	2026-02-11
312	Aditi Shah	212	Cancer Screening	2026-02-12
313	Dev Yadav	213	Kidney Infection	2026-02-13
314	Riya Jain	214	Asthma	2026-02-14
315	Yash Saini	215	Thyroid	2026-02-15
316	Mehak Khan	216	Ulcer	2026-02-16
317	Aryan Mishra	217	Arthritis	2026-02-17
318	Tanvi Agarwal	218	Appendicitis	2026-02-18
319	Kunal Malhotra	219	Heart Checkup	2026-02-19
320	Pari Kaur	220	Cold & Cough	2026-02-20
321	Om Vyas	221	Tooth Decay	2026-02-21
322	Diya Rao	222	Epilepsy	2026-02-22
323	Rudra Saxena	223	Knee Pain	2026-02-23
324	Saanvi Singh	224	High Fever	2026-02-24
325	Krish Bose	225	Acne	2026-02-25

```
25 rows in set (0.00 sec)
```

UPDATE

- Change the specialization of **Doctor ID 225** from **Dermatologist** to **Cosmetologist**.
- Update the experience of **Doctor ID 220** from **5 years** to **7 years**.
- Change the disease of **Patient ID 309** from **Anxiety** to **Depression**.
- Update the admission date of **Patient ID 320** to **2026-02-22**.
- Assign **Patient ID 323** to **Doctor ID 203**.

```
mysql> select * from Doctors;
```

doctor_id	doctor_name	specialization	experience
201	Dr. Amit Sharma	Cardiologist	15
202	Dr. Priya Verma	Neurologist	12
203	Dr. Rahul Singh	Orthopedic	10
204	Dr. Sneha Gupta	Pediatrician	8
205	Dr. Arjun Patel	Dermatologist	9
206	Dr. Neha Joshi	Gynecologist	14
207	Dr. Karan Mehta	ENT Specialist	11
208	Dr. Pooja Nair	Ophthalmologist	7
209	Dr. Rohan Das	Psychiatrist	13
210	Dr. Anjali Roy	General Physician	16
211	Dr. Vivek Kumar	Urologist	9
212	Dr. Meera Shah	Oncologist	17
213	Dr. Deepak Yadav	Nephrologist	12
214	Dr. Nisha Jain	Pulmonologist	10
215	Dr. Mohit Saini	Endocrinologist	8
216	Dr. Aisha Khan	Gastroenterologist	11
217	Dr. Saurabh Mishra	Rheumatologist	14
218	Dr. Ritu Agarwal	Surgeon	18
219	Dr. Akash Malhotra	Cardiologist	6
220	Dr. Simran Kaur	Pediatrician	7
221	Dr. Harsh Vyas	Dentist	9
222	Dr. Divya Rao	Neurologist	13
223	Dr. Manish Saxena	Orthopedic	7
224	Dr. Kavita Singh	General Physician	15
225	Dr. Aditya Bose	Cosmetologist	8

```
25 rows in set (0.00 sec)
```



```
mysql> select * from Patients;
```

patient_id	patient_name	doctor_id	disease	admission_date
301	Aarav Gupta	201	Heart Disease	2026-02-01
302	Siya Sharma	202	Migraine	2026-02-02
303	Rohan Verma	203	Fracture	2026-02-03
304	Ananya Singh	204	Viral Fever	2026-02-04
305	Kabir Patel	205	Skin Allergy	2026-02-05
306	Ishita Joshi	206	Pregnancy Checkup	2026-02-06
307	Vihaan Mehta	207	Ear Infection	2026-02-07
308	Myra Nair	208	Eye Infection	2026-02-08
309	Anjun Das	209	Depression	2026-02-09
310	Kiara Roy	210	Diabetes	2026-02-10
311	Aditya Kumar	211	Kidney Stone	2026-02-11
312	Aditi Shah	212	Cancer Screening	2026-02-12
313	Dev Yadav	213	Kidney Infection	2026-02-13
314	Riya Jain	214	Asthma	2026-02-14
315	Yash Saini	215	Thyroid	2026-02-15
316	Mehak Khan	216	Ulcer	2026-02-16
317	Aryan Mishra	217	Arthritis	2026-02-17
318	Tanvi Agarwal	218	Appendicitis	2026-02-18
319	Kunal Malhotra	219	Heart Checkup	2026-02-19
320	Pari Kaur	220	Cold & Cough	2026-02-22
321	Om Vyas	221	Tooth Decay	2026-02-21
322	Diya Rao	222	Epilepsy	2026-02-22
323	Rudra Saxena	203	Knee Pain	2026-02-23
324	Saanvi Singh	224	High Fever	2026-02-24
325	Krish Bose	225	Acne	2026-02-25

```
25 rows in set (0.00 sec)
```

DELETE

- a) Delete **Patient ID 325**.
- b) Delete **Doctor ID 225**.
- c) Delete the patient whose name is **Pari Kaur**.
- d) Delete the doctor whose name is **Dr. Harsh Vyas**.
- e) Delete the patient admitted on **2026-02-06**.

```
mysql> select * from Doctors;
```

doctor_id	doctor_name	specialization	experience
201	Dr. Amit Sharma	Cardiologist	15
202	Dr. Priya Verma	Neurologist	12
203	Dr. Rahul Singh	Orthopedic	10
204	Dr. Sneha Gupta	Pediatrician	8
205	Dr. Arjun Patel	Dermatologist	9
206	Dr. Neha Joshi	Gynecologist	14
207	Dr. Karan Mehta	ENT Specialist	11
208	Dr. Pooja Nair	Ophthalmologist	7
209	Dr. Rohan Das	Psychiatrist	13
210	Dr. Anjali Roy	General Physician	16
211	Dr. Vivek Kumar	Urologist	9
212	Dr. Meera Shah	Oncologist	17
213	Dr. Deepak Yadav	Nephrologist	12
214	Dr. Nisha Jain	Pulmonologist	10
215	Dr. Mohit Saini	Endocrinologist	8
216	Dr. Aisha Khan	Gastroenterologist	11
217	Dr. Saurabh Mishra	Rheumatologist	14
218	Dr. Ritu Agarwal	Surgeon	18
219	Dr. Akash Malhotra	Cardiologist	6
220	Dr. Simran Kaur	Pediatrician	7
222	Dr. Divya Rao	Neurologist	13
223	Dr. Manish Saxena	Orthopedic	7
224	Dr. Kavita Singh	General Physician	15

```
23 rows in set (0.00 sec)
```



```
mysql> select * from Patients;
```

patient_id	patient_name	doctor_id	disease	admission_date
301	Aarav Gupta	201	Heart Disease	2026-02-01
302	Siya Sharma	202	Migraine	2026-02-02
303	Rohan Verma	203	Fracture	2026-02-03
304	Ananya Singh	204	Viral Fever	2026-02-04
305	Kabir Patel	205	Skin Allergy	2026-02-05
307	Vihaan Mehta	207	Ear Infection	2026-02-07
308	Myra Nair	208	Eye Infection	2026-02-08
309	Arjun Das	209	Depression	2026-02-09
310	Kiara Roy	210	Diabetes	2026-02-10
311	Aditya Kumar	211	Kidney Stone	2026-02-11
312	Aditi Shah	212	Cancer Screening	2026-02-12
313	Dev Yadav	213	Kidney Infection	2026-02-13
314	Riya Jain	214	Asthma	2026-02-14
315	Yash Saini	215	Thyroid	2026-02-15
316	Mehak Khan	216	Ulcer	2026-02-16
317	Aryan Mishra	217	Arthritis	2026-02-17
318	Tanvi Agarwal	218	Appendicitis	2026-02-18
319	Kunal Malhotra	219	Heart Checkup	2026-02-19
322	Diya Rao	222	Epilepsy	2026-02-22
323	Rudra Saxena	203	Knee Pain	2026-02-23
324	Saanvi Singh	224	High Fever	2026-02-24

```
21 rows in set (0.00 sec)
```

JOIN

- a) Display patient names along with their doctor names.

```
mysql> SELECT
->   p.patient_name,
->   d.doctor_name
-> FROM Patients p
-> JOIN Doctors d
-> ON p.doctor_id=d.doctor_id;
```

patient_name	doctor_name
Aarav Gupta	Dr. Amit Sharma
Siya Sharma	Dr. Priya Verma
Rohan Verma	Dr. Rahul Singh
Ananya Singh	Dr. Sneha Gupta
Kabir Patel	Dr. Arjun Patel
Vihaan Mehta	Dr. Karan Mehta
Myra Nair	Dr. Pooja Nair
Arjun Das	Dr. Rohan Das
Kiara Roy	Dr. Anjali Roy
Aditya Kumar	Dr. Vivek Kumar
Aditi Shah	Dr. Meera Shah
Dev Yadav	Dr. Deepak Yadav
Riya Jain	Dr. Nisha Jain
Yash Saini	Dr. Mohit Saini
Mehak Khan	Dr. Aisha Khan
Aryan Mishra	Dr. Saurabh Mishra
Tanvi Agarwal	Dr. Ritu Agarwal
Kunal Malhotra	Dr. Akash Malhotra
Diya Rao	Dr. Divya Rao
Rudra Saxena	Dr. Rahul Singh
Saanvi Singh	Dr. Kavita Singh

21 rows in set (0.00 sec)

- b) Display patient name, disease, and doctor's specialization.

```
mysql> SELECT
->   p.patient_name,
->   p.disease,
->   d.specialization
-> FROM Patients p
-> JOIN Doctors d
-> ON p.doctor_id=d.doctor_id;
```

patient_name	disease	specialization
Aarav Gupta	Heart Disease	Cardiologist
Siya Sharma	Migraine	Neurologist
Rohan Verma	Fracture	Orthopedic
Ananya Singh	Viral Fever	Pediatrician
Kabir Patel	Skin Allergy	Dermatologist
Vihaan Mehta	Ear Infection	ENT Specialist
Myra Nair	Eye Infection	Ophthalmologist
Arjun Das	Depression	Psychiatrist
Kiara Roy	Diabetes	General Physician
Aditya Kumar	Kidney Stone	Urologist
Aditi Shah	Cancer Screening	Oncologist
Dev Yadav	Kidney Infection	Nephrologist
Riya Jain	Asthma	Pulmonologist
Yash Saini	Thyroid	Endocrinologist
Mehak Khan	Ulcer	Gastroenterologist
Aryan Mishra	Arthritis	Rheumatologist
Tanvi Agarwal	Appendicitis	Surgeon
Kunal Malhotra	Heart Checkup	Cardiologist
Diya Rao	Epilepsy	Neurologist
Rudra Saxena	Knee Pain	Orthopedic
Saanvi Singh	High Fever	General Physician

21 rows in set (0.00 sec)

- c) Display patient name, doctor name, and admission date.

```
mysql> SELECT
->   p.patient_name,
->   d.doctor_name,
->   p.admission_date
-> FROM Patients p
-> JOIN Doctors d
-> ON p.doctor_id=d.doctor_id;
```

patient_name	doctor_name	admission_date
Aarav Gupta	Dr. Amit Sharma	2026-02-01
Siya Sharma	Dr. Priya Verma	2026-02-02
Rohan Verma	Dr. Rahul Singh	2026-02-03
Ananya Singh	Dr. Sneha Gupta	2026-02-04
Kabir Patel	Dr. Arjun Patel	2026-02-05
Vihaan Mehta	Dr. Karan Mehta	2026-02-07
Myra Nair	Dr. Pooja Nair	2026-02-08
Arjun Das	Dr. Rohan Das	2026-02-09
Kiara Roy	Dr. Anjali Roy	2026-02-10
Aditya Kumar	Dr. Vivek Kumar	2026-02-11
Aditi Shah	Dr. Meera Shah	2026-02-12
Dev Yadav	Dr. Deepak Yadav	2026-02-13
Riya Jain	Dr. Nisha Jain	2026-02-14
Yash Saini	Dr. Mohit Saini	2026-02-15
Mehak Khan	Dr. Aisha Khan	2026-02-16
Aryan Mishra	Dr. Saurabh Mishra	2026-02-17
Tanvi Agarwal	Dr. Ritu Agarwal	2026-02-18
Kunal Malhotra	Dr. Akash Malhotra	2026-02-19
Diya Rao	Dr. Divya Rao	2026-02-22
Rudra Saxena	Dr. Rahul Singh	2026-02-23
Saanvi Singh	Dr. Kavita Singh	2026-02-24

21 rows in set (0.00 sec)

- d) Display doctor experience along with the patients assigned to them.

```
mysql> SELECT
->     d.doctor_name,
->     d.experience,
->     p.patient_name
-> FROM Doctors d
-> JOIN Patients p
-> ON d.doctor_id=p.doctor_id;
```

doctor_name	experience	patient_name
Dr. Amit Sharma	15	Aarav Gupta
Dr. Priya Verma	12	Siya Sharma
Dr. Rahul Singh	10	Rohan Verma
Dr. Sneha Gupta	8	Ananya Singh
Dr. Arjun Patel	9	Kabir Patel
Dr. Karan Mehta	11	Vihaan Mehta
Dr. Pooja Nair	7	Myra Nair
Dr. Rohan Das	13	Arjun Das
Dr. Anjali Roy	16	Kiara Roy
Dr. Vivek Kumar	9	Aditya Kumar
Dr. Meera Shah	17	Aditi Shah
Dr. Deepak Yadav	12	Dev Yadav
Dr. Nisha Jain	10	Riya Jain
Dr. Mohit Saini	8	Yash Saini
Dr. Aisha Khan	11	Mehak Khan
Dr. Saurabh Mishra	14	Aryan Mishra
Dr. Ritu Agarwal	18	Tanvi Agarwal
Dr. Akash Malhotra	6	Kunal Malhotra
Dr. Divya Rao	13	Diya Rao
Dr. Rahul Singh	10	Rudra Saxena
Dr. Kavita Singh	15	Saanvi Singh

21 rows in set (0.00 sec)

- e) Display all patients treated by **Cardiologists**.

```
mysql> SELECT
->     p.patient_name,
->     p.disease,
->     d.doctor_name
-> FROM Patients p
-> JOIN Doctors d
-> ON p.doctor_id=d.doctor_id
-> WHERE d.specialization='Cardiologist';
```

patient_name	disease	doctor_name
Aarav Gupta	Heart Disease	Dr. Amit Sharma
Kunal Malhotra	Heart Checkup	Dr. Akash Malhotra

2 rows in set (0.00 sec)

Problem 3: College Management System

Scenario

A college wants to maintain information about its departments and students. Each student belongs to one department. Students need to create the tables, insert the records, and practice **CREATE**, **INSERT**, **UPDATE**, **DELETE**, and **JOIN** operations.

Table 1: Departments

Department ID	Department Name	HOD Name	Building
301	Computer Science	Dr. Amit Sharma	Block A
302	Information Technology	Dr. Priya Verma	Block B
303	Electronics	Dr. Rahul Singh	Block C
304	Mechanical	Dr. Sneha Gupta	Block D

305	Civil	Dr. Arjun Patel	Block E
306	Electrical	Dr. Neha Joshi	Block F
307	Artificial Intelligence	Dr. Karan Mehta	Block G
308	Data Science	Dr. Pooja Nair	Block H
309	Biotechnology	Dr. Rohan Das	Block I
310	Chemical	Dr. Anjali Roy	Block J
311	Commerce	Dr. Vivek Kumar	Block K
312	Management	Dr. Meera Shah	Block L
313	Mathematics	Dr. Deepak Yadav	Block M
314	Physics	Dr. Nisha Jain	Block N
315	Chemistry	Dr. Mohit Saini	Block O
316	English	Dr. Aisha Khan	Block P
317	Economics	Dr. Saurabh Mishra	Block Q
318	Psychology	Dr. Ritu Agarwal	Block R
319	History	Dr. Akash Malhotra	Block S
320	Geography	Dr. Simran Kaur	Block T
321	Sociology	Dr. Harsh Vyas	Block U
322	Political Science	Dr. Divya Rao	Block V
323	Finance	Dr. Manish Saxena	Block W
324	Marketing	Dr. Kavita Singh	Block X
325	Human Resources	Dr. Aditya Bose	Block Y

Table 2: Students

Student ID	Student Name	Department ID	Semester	CGPA
401	Aarav Gupta	301	2	8.5
402	Siya Sharma	302	4	8.2
403	Rohan Verma	303	6	7.8
404	Ananya Singh	304	3	8.9
405	Kabir Patel	305	5	7.4
406	Ishita Joshi	306	1	9.1
407	Vihaan Mehta	307	7	8.7
408	Myra Nair	308	2	8.3
409	Arjun Das	309	4	7.6
410	Kiara Roy	310	8	8.8
411	Aditya Kumar	311	5	7.9
412	Aditi Shah	312	6	9.0
413	Dev Yadav	313	3	8.1
414	Riya Jain	314	2	7.5
415	Yash Saini	315	4	8.6
416	Mehak Khan	316	1	9.2
417	Aryan Mishra	317	7	8.0
418	Tanvi Agarwal	318	5	8.4
419	Kunal Malhotra	319	6	7.7
420	Pari Kaur	320	3	8.5
421	Om Vyas	321	2	7.9
422	Diya Rao	322	8	8.8
423	Rudra Saxena	323	4	8.1
424	Saanvi Singh	324	5	7.8
425	Krish Bose	325	6	9.3

Practice Questions

CREATE

- Create a database named **CollegeDB**
- Create the **Departments** table.
- Create the **Students** table with an appropriate foreign key.

```
mysql> CREATE DATABASE CollegeDB;
Query OK, 1 row affected (0.01 sec)

mysql>
mysql> USE CollegeDB;
Database changed
mysql> CREATE TABLE Departments(
  ->   department_id INT PRIMARY KEY,
  ->   department_name VARCHAR(50) NOT NULL,
  ->   hod_name VARCHAR(50) NOT NULL,
  ->   building VARCHAR(20) NOT NULL
  -> );
Query OK, 0 rows affected (0.04 sec)

mysql> CREATE TABLE Students(
  ->   student_id INT PRIMARY KEY,
  ->   student_name VARCHAR(50) NOT NULL,
  ->   department_id INT,
  ->   semester INT,
  ->   cgpa DECIMAL(3,1),
  ->   FOREIGN KEY(department_id)
  ->   REFERENCES Departments(department_id)
  -> );
Query OK, 0 rows affected (0.05 sec)

mysql> desc Departments;
+-----+-----+-----+-----+-----+-----+
| Field                | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| department_id        | int           | NO   | PRI | NULL    |       |
| department_name      | varchar(50)   | NO   |     | NULL    |       |
| hod_name             | varchar(50)   | NO   |     | NULL    |       |
| building             | varchar(20)   | NO   |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.01 sec)

mysql> desc Students;
+-----+-----+-----+-----+-----+-----+
| Field                | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| student_id           | int           | NO   | PRI | NULL    |       |
| student_name         | varchar(50)   | NO   |     | NULL    |       |
| department_id        | int           | YES  | MUL | NULL    |       |
| semester             | int           | YES  |     | NULL    |       |
| cgpa                 | decimal(3,1)  | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

INSERT

- Insert all 25 department records.

```
mysql> select * from Departments;
+-----+-----+-----+-----+
| department_id | department_name | hod_name | building |
+-----+-----+-----+-----+
| 301           | Computer Science | Dr. Amit Sharma | Block A |
| 302           | Information Technology | Dr. Priya Verma | Block B |
| 303           | Electronics      | Dr. Rahul Singh | Block C |
| 304           | Mechanical       | Dr. Sneha Gupta | Block D |
| 305           | Civil            | Dr. Arjun Patel | Block E |
| 306           | Electrical       | Dr. Neha Joshi | Block F |
| 307           | Artificial Intelligence | Dr. Karan Mehta | Block G |
| 308           | Data Science    | Dr. Pooja Nair | Block H |
| 309           | Biotechnology   | Dr. Rohan Das | Block I |
| 310           | Chemical        | Dr. Anjali Roy | Block J |
| 311           | Commerce        | Dr. Vivek Kumar | Block K |
| 312           | Management      | Dr. Meera Shah | Block L |
| 313           | Mathematics     | Dr. Deepak Yadav | Block M |
| 314           | Physics         | Dr. Nisha Jain | Block N |
| 315           | Chemistry       | Dr. Mohit Saini | Block O |
| 316           | English         | Dr. Aisha Khan | Block P |
| 317           | Economics       | Dr. Saurabh Mishra | Block Q |
| 318           | Psychology      | Dr. Ritu Agarwal | Block R |
| 319           | History         | Dr. Akash Malhotra | Block S |
| 320           | Geography       | Dr. Simran Kaur | Block T |
| 321           | Sociology       | Dr. Harsh Vyas | Block U |
| 322           | Political Science | Dr. Divya Rao | Block V |
| 323           | Finance         | Dr. Manish Saxena | Block W |
| 324           | Marketing       | Dr. Kavita Singh | Block X |
| 325           | Human Resources | Dr. Aditya Bose | Block Y |
+-----+-----+-----+-----+
25 rows in set (0.00 sec)
```

b) Insert all 25 student records.

```
mysql> select * from Students;
```

student_id	student_name	department_id	semester	cgpa
401	Aarav Gupta	301	2	8.5
402	Siya Sharma	302	4	8.2
403	Rohan Verma	303	6	7.8
404	Ananya Singh	304	3	8.9
405	Kabir Patel	305	5	7.4
406	Ishita Joshi	306	1	9.1
407	Vihaan Mehta	307	7	8.7
408	Myra Nair	308	2	8.3
409	Arjun Das	309	4	7.6
410	Kiara Roy	310	8	8.8
411	Aditya Kumar	311	5	7.9
412	Aditi Shah	312	6	9.0
413	Dev Yadav	313	3	8.1
414	Riya Jain	314	2	7.5
415	Yash Saini	315	4	8.6
416	Mehak Khan	316	1	9.2
417	Aryan Mishra	317	7	8.0
418	Tanvi Agarwal	318	5	8.4
419	Kunal Malhotra	319	6	7.7
420	Pari Kaur	320	3	8.5
421	Om Vyas	321	2	7.9
422	Diya Rao	322	8	8.8
423	Rudra Saxena	323	4	8.1
424	Saanvi Singh	324	5	7.8
425	Krish Bose	325	6	9.3

```
25 rows in set (0.00 sec)
```

UPDATE

- Change the **HOD** of the **Artificial Intelligence** department to **Dr. Rakesh Gupta**.
- Update the **Building** of the **Marketing** department from **Block X** to **Block Z**.
- Change the **CGPA** of **Student ID 417** from **8.0** to **8.6**.
- Update the **Semester** of **Student ID 406** from **1** to **2**.
- Transfer **Student ID 424** from the **Marketing** department to the **Finance** Department.

```
mysql> UPDATE Departments
-> SET hod_name='Dr. Rakesh Gupta'
-> WHERE department_name='Artificial Intelligence';
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> UPDATE Departments
-> SET building='Block Z'
-> WHERE department_name='Marketing';
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> UPDATE Students
-> SET cgpa=8.6
-> WHERE student_id=417;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> UPDATE Students
-> SET semester=2
-> WHERE student_id=406;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> UPDATE Students
-> SET department_id=323
-> WHERE student_id=424;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0
```

```
mysql> select *from Departments;
```

department_id	department_name	hod_name	building
301	Computer Science	Dr. Amit Sharma	Block A
302	Information Technology	Dr. Priya Verma	Block B
303	Electronics	Dr. Rahul Singh	Block C
304	Mechanical	Dr. Sneha Gupta	Block D
305	Civil	Dr. Arjun Patel	Block E
306	Electrical	Dr. Neha Joshi	Block F
307	Artificial Intelligence	Dr. Rakesh Gupta	Block G
308	Data Science	Dr. Pooja Nair	Block H
309	Biotechnology	Dr. Rohan Das	Block I
310	Chemical	Dr. Anjali Roy	Block J
311	Commerce	Dr. Vivek Kumar	Block K
312	Management	Dr. Meera Shah	Block L
313	Mathematics	Dr. Deepak Yadav	Block M
314	Physics	Dr. Nisha Jain	Block N
315	Chemistry	Dr. Mohit Saini	Block O
316	English	Dr. Aisha Khan	Block P
317	Economics	Dr. Saurabh Mishra	Block Q
318	Psychology	Dr. Ritu Agarwal	Block R
319	History	Dr. Akash Malhotra	Block S
320	Geography	Dr. Simran Kaur	Block T
321	Sociology	Dr. Harsh Vyas	Block U
322	Political Science	Dr. Divya Rao	Block V
323	Finance	Dr. Manish Saxena	Block W
324	Marketing	Dr. Kavita Singh	Block Z
325	Human Resources	Dr. Aditya Bose	Block Y

```
25 rows in set (0.00 sec)
```

```
mysql> select *from Students;
```

student_id	student_name	department_id	semester	cgpa
401	Aarav Gupta	301	2	8.5
402	Siya Sharma	302	4	8.2
403	Rohan Verma	303	6	7.8
404	Ananya Singh	304	3	8.9
405	Kabir Patel	305	5	7.4
406	Ishita Joshi	306	2	9.1
407	Vihaan Mehta	307	7	8.7
408	Myra Nair	308	2	8.3
409	Arjun Das	309	4	7.6
410	Kiara Roy	310	8	8.8
411	Aditya Kumar	311	5	7.9
412	Aditi Shah	312	6	9.0
413	Dev Yadav	313	3	8.1
414	Riya Jain	314	2	7.5
415	Yash Saini	315	4	8.6
416	Mehak Khan	316	1	9.2
417	Aryan Mishra	317	7	8.6
418	Tanvi Agarwal	318	5	8.4
419	Kunal Malhotra	319	6	7.7
420	Pari Kaur	320	3	8.5
421	Om Vyas	321	2	7.9
422	Diya Rao	322	8	8.8
423	Rudra Saxena	323	4	8.1
424	Saanvi Singh	323	5	7.8
425	Krish Bose	325	6	9.3

```
25 rows in set (0.00 sec)
```

DELETE

- a) Delete **Student ID 425**.
- b) Delete the **Human Resources** department.
- c) Delete the student named **Pari Kaur**.
- d) Delete the department named **History**.
- e) Delete the student with **Student ID 411**.

```
mysql> select * from Departments;
```

department_id	department_name	hod_name	building
301	Computer Science	Dr. Amit Sharma	Block A
302	Information Technology	Dr. Priya Verma	Block B
303	Electronics	Dr. Rahul Singh	Block C
304	Mechanical	Dr. Sneha Gupta	Block D
305	Civil	Dr. Arjun Patel	Block E
306	Electrical	Dr. Neha Joshi	Block F
307	Artificial Intelligence	Dr. Rakesh Gupta	Block G
308	Data Science	Dr. Pooja Nair	Block H
309	Biotechnology	Dr. Rohan Das	Block I
310	Chemical	Dr. Anjali Roy	Block J
311	Commerce	Dr. Vivek Kumar	Block K
312	Management	Dr. Meera Shah	Block L
313	Mathematics	Dr. Deepak Yadav	Block M
314	Physics	Dr. Nisha Jain	Block N
315	Chemistry	Dr. Mohit Saini	Block O
316	English	Dr. Aisha Khan	Block P
317	Economics	Dr. Saurabh Mishra	Block Q
318	Psychology	Dr. Ritu Agarwal	Block R
320	Geography	Dr. Simran Kaur	Block T
321	Sociology	Dr. Harsh Vyas	Block U
322	Political Science	Dr. Divya Rao	Block V
323	Finance	Dr. Manish Saxena	Block W
324	Marketing	Dr. Kavita Singh	Block Z

```
23 rows in set (0.00 sec)
```

```
mysql> select * from Students;
```

student_id	student_name	department_id	semester	cgpa
401	Aarav Gupta	301	2	8.5
402	Siya Sharma	302	4	8.2
403	Rohan Verma	303	6	7.8
404	Ananya Singh	304	3	8.9
405	Kabir Patel	305	5	7.4
406	Ishita Joshi	306	2	9.1
407	Vihaan Mehta	307	7	8.7
408	Myra Nair	308	2	8.3
409	Arjun Das	309	4	7.6
410	Kiara Roy	310	8	8.8
412	Aditi Shah	312	6	9.0
413	Dev Yadav	313	3	8.1
414	Riya Jain	314	2	7.5
415	Yash Saini	315	4	8.6
416	Mehak Khan	316	1	9.2
417	Aryan Mishra	317	7	8.6
418	Tanvi Agarwal	318	5	8.4
421	Om Vyas	321	2	7.9
422	Diya Rao	322	8	8.8
423	Rudra Saxena	323	4	8.1
424	Saanvi Singh	323	5	7.8

```
21 rows in set (0.00 sec)
```

JOIN

- a) Display the **student name** along with their **department name**.

```
mysql> SELECT
->     s.student_name,
->     d.department_name
-> FROM Students s
-> JOIN Departments d
-> ON s.department_id=d.department_id;
```

student_name	department_name
Aarav Gupta	Computer Science
Siya Sharma	Information Technology
Rohan Verma	Electronics
Ananya Singh	Mechanical
Kabir Patel	Civil
Ishita Joshi	Electrical
Vihaan Mehta	Artificial Intelligence
Myra Nair	Data Science
Arjun Das	Biotechnology
Kiara Roy	Chemical
Aditi Shah	Management
Dev Yadav	Mathematics
Riya Jain	Physics
Yash Saini	Chemistry
Mehak Khan	English
Aryan Mishra	Economics
Tanvi Agarwal	Psychology
Om Vyas	Sociology
Diya Rao	Political Science
Rudra Saxena	Finance
Saanvi Singh	Finance

21 rows in set (0.00 sec)

- b) Display the **student name**, **department name**, and **HOD name**.

```
mysql> SELECT
->     s.student_name,
->     d.department_name,
->     d.hod_name
-> FROM Students s
-> JOIN Departments d
-> ON s.department_id=d.department_id;
```

student_name	department_name	hod_name
Aarav Gupta	Computer Science	Dr. Amit Sharma
Siya Sharma	Information Technology	Dr. Priya Verma
Rohan Verma	Electronics	Dr. Rahul Singh
Ananya Singh	Mechanical	Dr. Sneha Gupta
Kabir Patel	Civil	Dr. Arjun Patel
Ishita Joshi	Electrical	Dr. Neha Joshi
Vihaan Mehta	Artificial Intelligence	Dr. Rakesh Gupta
Myra Nair	Data Science	Dr. Pooja Nair
Arjun Das	Biotechnology	Dr. Rohan Das
Kiara Roy	Chemical	Dr. Anjali Roy
Aditi Shah	Management	Dr. Meera Shah
Dev Yadav	Mathematics	Dr. Deepak Yadav
Riya Jain	Physics	Dr. Nisha Jain
Yash Saini	Chemistry	Dr. Mohit Saini
Mehak Khan	English	Dr. Aisha Khan
Aryan Mishra	Economics	Dr. Saurabh Mishra
Tanvi Agarwal	Psychology	Dr. Ritu Agarwal
Om Vyas	Sociology	Dr. Harsh Vyas
Diya Rao	Political Science	Dr. Divya Rao
Rudra Saxena	Finance	Dr. Manish Saxena
Saanvi Singh	Finance	Dr. Manish Saxena

21 rows in set (0.00 sec)

- c) Display the **student name**, **semester**, and **building** of their department.

```
mysql> SELECT
->     s.student_name,
->     s.semester,
->     d.building
-> FROM Students s
-> JOIN Departments d
-> ON s.department_id=d.department_id;
```

student_name	semester	building
Aarav Gupta	2	Block A
Siya Sharma	4	Block B
Rohan Verma	6	Block C
Ananya Singh	3	Block D
Kabir Patel	5	Block E
Ishita Joshi	2	Block F
Vihaan Mehta	7	Block G
Myra Nair	2	Block H
Arjun Das	4	Block I
Kiara Roy	8	Block J
Aditi Shah	6	Block L
Dev Yadav	3	Block M
Riya Jain	2	Block N
Yash Saini	4	Block O
Mehak Khan	1	Block P
Aryan Mishra	7	Block Q
Tanvi Agarwal	5	Block R
Om Vyas	2	Block U
Diya Rao	8	Block V
Rudra Saxena	4	Block W
Saanvi Singh	5	Block W

21 rows in set (0.00 sec)

- d) Display all students who belong to the **Computer Science** department.

```
mysql> SELECT
->     s.student_name,
->     d.department_name
-> FROM Students s
-> JOIN Departments d
-> ON s.department_id=d.department_id
-> WHERE d.department_name='Computer Science';
```

student_name	department_name
Aarav Gupta	Computer Science

1 row in set (0.00 sec)

- e) Display the **student name**, **department name**, and **CGPA**.

```
mysql> SELECT
->     s.student_name,
->     d.department_name,
->     s.cgpa
-> FROM Students s
-> JOIN Departments d
-> ON s.department_id=d.department_id;
```

student_name	department_name	cgpa
Aarav Gupta	Computer Science	8.5
Siya Sharma	Information Technology	8.2
Rohan Verma	Electronics	7.8
Ananya Singh	Mechanical	8.9
Kabir Patel	Civil	7.4
Ishita Joshi	Electrical	9.1
Vihaan Mehta	Artificial Intelligence	8.7
Myra Nair	Data Science	8.3
Arjun Das	Biotechnology	7.6
Kiara Roy	Chemical	8.8
Aditi Shah	Management	9.0
Dev Yadav	Mathematics	8.1
Riya Jain	Physics	7.5
Yash Saini	Chemistry	8.6
Mehak Khan	English	9.2
Aryan Mishra	Economics	8.6
Tanvi Agarwal	Psychology	8.4
Om Vyas	Sociology	7.9
Diya Rao	Political Science	8.8
Rudra Saxena	Finance	8.1
Saanvi Singh	Finance	7.8

21 rows in set (0.00 sec)

Problem 4: Movie Ticket Booking System

Scenario

A multiplex cinema wants to maintain information about the movies currently running and the tickets booked by customers. Each booking is associated with one movie. Students need to create the tables, insert the records, and practice **CREATE**, **INSERT**, **UPDATE**, **DELETE**, and **JOIN** operations.

Table 1: Movies

Movie ID	Movie Name	Language	Genre	Ticket Price (₹)
401	Pushpa 3	Telugu	Action	350
402	Kalki 2898 AD	Telugu	Sci-Fi	400
403	Chhaava	Hindi	Historical	300
404	Housefull 5	Hindi	Comedy	250
405	Coolie	Tamil	Action	320
406	Jailer 2	Tamil	Action	340
407	Kantara Chapter 1	Kannada	Thriller	330
408	Martin	Kannada	Action	290
409	Sitaare Zameen Par	Hindi	Drama	280
410	War 2	Hindi	Action	380
411	Dragon	Tamil	Drama	260

412	HIT 3	Telugu	Crime	310
413	Tourist Family	Tamil	Family	220
414	Retro	Tamil	Romance	240
415	Thug Life	Tamil	Action	360
416	Lucifer 2	Malayalam	Action	330
417	Empuraan	Malayalam	Thriller	350
418	Marco	Malayalam	Crime	300
419	Deva	Hindi	Action	290
420	Bhool Chuk Maaf	Hindi	Comedy	240
421	Kanguva	Tamil	Fantasy	370
422	Vidaamuyarchi	Tamil	Thriller	320
423	Game Changer	Telugu	Political Drama	340
424	Indian 3	Tamil	Action	360
425	Baaghi 4	Hindi	Action	310

Table 2: Bookings

Booking ID	Customer Name	Movie ID	Number of Tickets	Booking Date
501	Aarav Gupta	401	2	2026-03-01
502	Siya Sharma	402	4	2026-03-01
503	Rohan Verma	403	3	2026-03-02
504	Ananya Singh	404	2	2026-03-02
505	Kabir Patel	405	5	2026-03-03
506	Ishita Joshi	406	2	2026-03-03
507	Vihaan Mehta	407	1	2026-03-04
508	Myra Nair	408	3	2026-03-04
509	Arjun Das	409	2	2026-03-05
510	Kiara Roy	410	4	2026-03-05
511	Aditya Kumar	411	2	2026-03-06
512	Aditi Shah	412	3	2026-03-06
513	Dev Yadav	413	4	2026-03-07
514	Riya Jain	414	2	2026-03-07
515	Yash Saini	415	5	2026-03-08
516	Mehak Khan	416	1	2026-03-08
517	Aryan Mishra	417	3	2026-03-09
518	Tanvi Agarwal	418	2	2026-03-09
519	Kunal Malhotra	419	4	2026-03-10
520	Pari Kaur	420	2	2026-03-10
521	Om Vyas	421	5	2026-03-11
522	Diya Rao	422	3	2026-03-11
523	Rudra Saxena	423	2	2026-03-12
524	Saanvi Singh	424	4	2026-03-12
525	Krish Bose	425	1	2026-03-13

Practice Questions

CREATE

- Create a database named **MovieBookingDB**.
- Create the **Movies** table.
- Create the **Bookings** table with an appropriate foreign key.

```
mysql> CREATE DATABASE MovieBookingDB;
Query OK, 1 row affected (0.01 sec)

mysql>
mysql> USE MovieBookingDB;
Database changed
mysql> CREATE TABLE Movies(
->     movie_id INT PRIMARY KEY,
->     movie_name VARCHAR(100) NOT NULL,
->     language VARCHAR(30) NOT NULL,
->     genre VARCHAR(50) NOT NULL,
->     ticket_price DECIMAL(6,2) NOT NULL
-> );
Query OK, 0 rows affected (0.03 sec)

mysql> CREATE TABLE Bookings(
->     booking_id INT PRIMARY KEY,
->     customer_name VARCHAR(50) NOT NULL,
->     movie_id INT,
->     number_of_tickets INT,
->     booking_date DATE,
->     FOREIGN KEY(movie_id)
->     REFERENCES Movies(movie_id)
-> );
Query OK, 0 rows affected (0.09 sec)

mysql> desc Movies;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| movie_id   | int           | NO   | PRI | NULL    |       |
| movie_name | varchar(100)  | NO   |     | NULL    |       |
| language   | varchar(30)   | NO   |     | NULL    |       |
| genre      | varchar(50)   | NO   |     | NULL    |       |
| ticket_price | decimal(6,2) | NO   |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> desc Bookings;
+-----+-----+-----+-----+-----+-----+
| Field              | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| booking_id         | int           | NO   | PRI | NULL    |       |
| customer_name      | varchar(50)   | NO   |     | NULL    |       |
| movie_id           | int           | YES  | MUL | NULL    |       |
| number_of_tickets  | int           | YES  |     | NULL    |       |
| booking_date       | date          | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

INSERT

- Insert all 25 movie records.

```
mysql> select * from Movies;
+-----+-----+-----+-----+-----+
| movie_id | movie_name      | language | genre      | ticket_price |
+-----+-----+-----+-----+-----+
| 401      | Pushpa 3        | Telugu   | Action     | 350.00       |
| 402      | Kalki 2898 AD  | Telugu   | Sci-Fi     | 400.00       |
| 403      | Chhaava         | Hindi    | Historical  | 300.00       |
| 404      | Housefull 5     | Hindi    | Comedy     | 250.00       |
| 405      | Coolie          | Tamil    | Action     | 320.00       |
| 406      | Jailer 2        | Tamil    | Action     | 340.00       |
| 407      | Kantara Chapter 1 | Kannada  | Thriller   | 330.00       |
| 408      | Martin          | Kannada  | Action     | 290.00       |
| 409      | Sitaare Zameen Par | Hindi    | Drama      | 280.00       |
| 410      | War 2           | Hindi    | Action     | 380.00       |
| 411      | Dragon          | Tamil    | Drama      | 260.00       |
| 412      | HIT 3           | Telugu   | Crime      | 310.00       |
| 413      | Tourist Family  | Tamil    | Family     | 220.00       |
| 414      | Retro           | Tamil    | Romance    | 240.00       |
| 415      | Thug Life       | Tamil    | Action     | 360.00       |
| 416      | Lucifer 2       | Malayalam | Action     | 330.00       |
| 417      | Empuraan        | Malayalam | Thriller   | 350.00       |
| 418      | Marco           | Malayalam | Crime      | 300.00       |
| 419      | Deva            | Hindi    | Action     | 290.00       |
| 420      | Bhool Chuk Maaf | Hindi    | Comedy     | 240.00       |
| 421      | Kanguva         | Tamil    | Fantasy    | 370.00       |
| 422      | Vidaamuyarchi   | Tamil    | Thriller   | 320.00       |
| 423      | Game Changer    | Telugu   | Political Drama | 340.00       |
| 424      | Indian 3        | Tamil    | Action     | 360.00       |
| 425      | Baaghi 4        | Hindi    | Action     | 310.00       |
+-----+-----+-----+-----+-----+
25 rows in set (0.00 sec)
```

b) Insert all 25 booking records.

```
mysql> select * from Bookings;
```

booking_id	customer_name	movie_id	number_of_tickets	booking_date
501	Aarav Gupta	401	2	2026-03-01
502	Siya Sharma	402	4	2026-03-01
503	Rohan Verma	403	3	2026-03-02
504	Ananya Singh	404	2	2026-03-02
505	Kabir Patel	405	2	2026-03-03
506	Ishita Joshi	406	2	2026-03-03
507	Vihaan Mehta	407	1	2026-03-04
508	Myra Nair	408	3	2026-03-04
509	Arjun Das	409	2	2026-03-05
510	Kiara Roy	410	4	2026-03-05
511	Aditya Kumar	411	2	2026-03-06
512	Aditi Shah	412	3	2026-03-06
513	Dev Yadav	413	4	2026-03-07
514	Riya Jain	414	2	2026-03-07
515	Yash Saini	415	2	2026-03-08
516	Mehak Khan	416	3	2026-03-08
517	Aryan Mishra	417	2	2026-03-09
518	Tanvi Agarwal	418	2	2026-03-09
519	Kunal Malhotra	419	4	2026-03-10
520	Pari Kaur	420	2	2026-03-10
521	Om Vyas	421	3	2026-03-11
522	Diya Rao	422	2	2026-03-11
523	Rudra Saxena	423	4	2026-03-12
524	Saanvi Singh	424	2	2026-03-12
525	Krish Bose	425	1	2026-03-13

25 rows in set (0.00 sec)

UPDATE

- Change the ticket price of **War 2** from ₹380 to ₹400.
- Update the genre of **Housefull 5** from **Comedy** to **Comedy Drama**.
- Change the number of tickets booked by **Siya Sharma** from 4 to 5.
- Update the booking date of **Booking ID 516** to 2026-03-10.
- Change the movie booked by **Krish Bose** from **Baaghi 4** to **Pushpa 3**.

```
mysql> UPDATE Movies
-> SET ticket_price=400
-> WHERE movie_name='War 2';
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 changed: 1 warnings: 0

mysql> UPDATE Movies
-> SET genre='Comedy Drama'
-> WHERE movie_name='Housefull 5';
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 changed: 1 warnings: 0

mysql> UPDATE Bookings
-> SET number_of_tickets=5
-> WHERE customer_name='Siya Sharma';
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 changed: 1 warnings: 0

mysql> UPDATE Bookings
-> SET booking_date='2026-03-10'
-> WHERE booking_id=516;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 changed: 1 warnings: 0

mysql> UPDATE Bookings
-> SET movie_id=401
-> WHERE booking_id=525;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 changed: 1 warnings: 0
```

```
mysql> select * from Movies;
```

movie_id	movie_name	language	genre	ticket_price
401	Pushpa 3	Telugu	Action	350.00
402	Kalki 2898 AD	Telugu	Sci-Fi	400.00
403	Chhaava	Hindi	Historical	300.00
404	Housefull 5	Hindi	Comedy Drama	250.00
405	Coolie	Tamil	Action	320.00
406	Jailer 2	Tamil	Action	340.00
407	Kantara Chapter 1	Kannada	Thriller	390.00
408	Martin	Kannada	Action	390.00
409	Sitaare Zameen Par	Hindi	Drama	280.00
410	War 2	Hindi	Action	400.00
411	Dragon	Tamil	Drama	260.00
412	HIT 3	Telugu	Crime	320.00
413	Tourist Family	Tamil	Family	220.00
414	Retro	Tamil	Romance	240.00
415	Thug Life	Tamil	Action	360.00
416	Lucifer 2	Malayalam	Action	330.00
417	Empuraan	Malayalam	Thriller	350.00
418	Marco	Malayalam	Crime	300.00
419	Deva	Hindi	Action	290.00
420	Bhool Chuk Maaf	Hindi	Comedy	240.00
421	Kanguva	Tamil	Fantasy	370.00
422	Vidaamuyarchi	Tamil	Thriller	320.00
423	Game Changer	Telugu	Political Drama	340.00
424	Indian 3	Tamil	Action	360.00
425	Baaghi 4	Hindi	Action	310.00

25 rows in set (0.00 sec)

```
mysql> select * from Bookings;
```

booking_id	customer_name	movie_id	number_of_tickets	booking_date
501	Aarav Gupta	401	2	2026-03-01
502	Siya Sharma	402	5	2026-03-01
503	Rohan Verma	403	3	2026-03-02
504	Ananya Singh	404	2	2026-03-02
505	Kabir Patel	405	2	2026-03-03
506	Ishita Joshi	406	2	2026-03-03
507	Vihaan Mehta	407	1	2026-03-04
508	Myra Nair	408	3	2026-03-04
509	Arjun Das	409	2	2026-03-05
510	Kiara Roy	410	4	2026-03-05
511	Aditya Kumar	411	2	2026-03-06
512	Aditi Shah	412	3	2026-03-06
513	Dev Yadav	413	4	2026-03-07
514	Riya Jain	414	2	2026-03-07
515	Yash Saini	415	2	2026-03-08
516	Mehak Khan	416	3	2026-03-08
517	Aryan Mishra	417	2	2026-03-09
518	Tanvi Agarwal	418	2	2026-03-09
519	Kunal Malhotra	419	4	2026-03-10
520	Pari Kaur	420	2	2026-03-10
521	Om Vyas	421	3	2026-03-11
522	Diya Rao	422	2	2026-03-11
523	Rudra Saxena	423	4	2026-03-12
524	Saanvi Singh	424	2	2026-03-12
525	Krish Bose	401	1	2026-03-13

25 rows in set (0.00 sec)

DELETE

- Delete **Booking ID 525**.
- Delete the movie **Indian 3**.
- Delete the booking made by **Vihaan Mehta**.
- Delete the movie **Dragon**.
- Delete the booking made on **2026-03-03** by **Kabir Patel**.

```
mysql> select * from Movies;
```

movie_id	movie_name	language	genre	ticket_price
401	Pushpa 3	Telugu	Action	350.00
402	Kalki 2898 AD	Telugu	Sci-Fi	400.00
403	Chhaava	Hindi	Historical	300.00
404	Housefull 5	Hindi	Comedy Drama	250.00
405	Coolie	Tamil	Action	320.00
406	Jailer 2	Tamil	Action	340.00
407	Kantara Chapter 1	Kannada	Thriller	330.00
408	Martin	Kannada	Action	290.00
409	Sitaare Zameen Par	Hindi	Drama	280.00
410	War 2	Hindi	Action	400.00
412	HIT 3	Telugu	Crime	310.00
413	Tourist Family	Tamil	Family	220.00
414	Retro	Tamil	Romance	240.00
415	Thug Life	Tamil	Action	360.00
416	Lucifer 2	Malayalam	Action	330.00
417	Empuraan	Malayalam	Thriller	350.00
418	Marco	Malayalam	Crime	300.00
419	Deva	Hindi	Action	290.00
420	Bhool Chuk Maaf	Hindi	Comedy	240.00
421	Kanguva	Tamil	Fantasy	370.00
422	Vidaamuyarchi	Tamil	Thriller	320.00
423	Game Changer	Telugu	Political Drama	340.00
425	Baaghi 4	Hindi	Action	310.00

23 rows in set (0.00 sec)

```
mysql> select * from Bookings;
```

booking_id	customer_name	movie_id	number_of_tickets	booking_date
501	Aarav Gupta	401	2	2026-03-01
502	Siya Sharma	402	5	2026-03-01
503	Rohan Verma	403	3	2026-03-02
504	Ananya Singh	404	2	2026-03-02
506	Ishita Joshi	406	2	2026-03-03
508	Myra Nair	408	3	2026-03-04
509	Arjun Das	409	2	2026-03-05
510	Kiara Roy	410	4	2026-03-05
512	Aditi Shah	412	3	2026-03-06
513	Dev Yadav	413	4	2026-03-07
514	Riya Jain	414	2	2026-03-07
515	Yash Saini	415	5	2026-03-08
516	Mehak Khan	416	1	2026-03-10
517	Aryan Mishra	417	3	2026-03-09
518	Tanvi Agarwal	418	2	2026-03-09
519	Kunal Malhotra	419	4	2026-03-10
520	Pari Kaur	420	2	2026-03-10
521	Om Vyas	421	5	2026-03-11
522	Diya Rao	422	3	2026-03-11
523	Rudra Saxena	423	2	2026-03-12

20 rows in set (0.00 sec)

JOIN

- a) Display the **customer name** along with the **movie name** they booked.

```
mysql> SELECT
->     b.customer_name,
->     m.movie_name
-> FROM Bookings b
-> JOIN Movies m
-> ON b.movie_id=m.movie_id;
```

customer_name	movie_name
Aarav Gupta	Pushpa 3
Siya Sharma	Kalki 2898 AD
Rohan Verma	Chhaava
Ananya Singh	Housefull 5
Ishita Joshi	Jailer 2
Myra Nair	Martin
Arjun Das	Sitaare Zameen Par
Kiara Roy	War 2
Aditi Shah	HIT 3
Dev Yadav	Tourist Family
Riya Jain	Retro
Yash Saini	Thug Life
Mehak Khan	Lucifer 2
Aryan Mishra	Empuraan
Tanvi Agarwal	Marco
Kunal Malhotra	Deva
Pari Kaur	Bhool Chuk Maaf
Om Vyas	Kanguva
Diya Rao	Vidaamuyarchi
Rudra Saxena	Game Changer

20 rows in set (0.00 sec)

- b) Display the **customer name**, **movie language**, and **booking date**.

```
mysql> SELECT
->     b.customer_name,
->     m.language,
->     b.booking_date
-> FROM Bookings b
-> JOIN Movies m
-> ON b.movie_id=m.movie_id;
```

customer_name	language	booking_date
Aarav Gupta	Telugu	2026-03-01
Siya Sharma	Telugu	2026-03-01
Rohan Verma	Hindi	2026-03-02
Ananya Singh	Hindi	2026-03-02
Ishita Joshi	Tamil	2026-03-03
Myra Nair	Kannada	2026-03-04
Arjun Das	Hindi	2026-03-05
Kiara Roy	Hindi	2026-03-05
Aditi Shah	Telugu	2026-03-06
Dev Yadav	Tamil	2026-03-07
Riya Jain	Tamil	2026-03-07
Yash Saini	Tamil	2026-03-08
Mehak Khan	Malayalam	2026-03-10
Aryan Mishra	Malayalam	2026-03-09
Tanvi Agarwal	Malayalam	2026-03-09
Kunal Malhotra	Hindi	2026-03-10
Pari Kaur	Hindi	2026-03-10
Om Vyas	Tamil	2026-03-11
Diya Rao	Tamil	2026-03-11
Rudra Saxena	Telugu	2026-03-12

20 rows in set (0.00 sec)

- c) Display the **customer name**, **movie genre**, and **number of tickets** booked.

```
mysql> SELECT
->     b.customer_name,
->     m.genre,
->     b.number_of_tickets
-> FROM Bookings b
-> JOIN Movies m
-> ON b.movie_id=m.movie_id;
```

customer_name	genre	number_of_tickets
Aarav Gupta	Action	2
Siya Sharma	Sci-Fi	5
Rohan Verma	Historical	3
Ananya Singh	Comedy Drama	2
Ishita Joshi	Action	2
Myra Nair	Action	3
Arjun Das	Drama	2
Kiara Roy	Action	4
Aditi Shah	Crime	3
Dev Yadav	Family	4
Riya Jain	Romance	2
Yash Saini	Action	5
Mehak Khan	Action	1
Aryan Mishra	Thriller	3
Tanvi Agarwal	Crime	2
Kunal Malhotra	Action	4
Pari Kaur	Comedy	2
Om Vyas	Fantasy	5
Diya Rao	Thriller	3
Rudra Saxena	Political Drama	2

20 rows in set (0.00 sec)

- d) Display all customers who booked **Hindi** movies.

```
mysql> SELECT
->     b.customer_name,
->     m.movie_name
-> FROM Bookings b
-> JOIN Movies m
-> ON b.movie_id=m.movie_id
-> WHERE m.language='Hindi';
```

customer_name	movie_name
Rohan Verma	Chhaava
Ananya Singh	Housefull 5
Arjun Das	Sitaare Zameen Par
Kiara Roy	War 2
Kunal Malhotra	Deva
Pari Kaur	Bhool Chuk Maaf

6 rows in set (0.00 sec)

- e) Display the **customer name**, **movie name**, and **ticket price**.

```
mysql> SELECT
->     b.customer_name,
->     m.movie_name,
->     m.ticket_price
-> FROM Bookings b
-> JOIN Movies m
-> ON b.movie_id=m.movie_id;
```

customer_name	movie_name	ticket_price
Aarav Gupta	Pushpa 3	350.00
Siya Sharma	Kalki 2898 AD	400.00
Rohan Verma	Chhaava	300.00
Ananya Singh	Housefull 5	250.00
Ishita Joshi	Jailer 2	340.00
Myra Nair	Martin	290.00
Arjun Das	Sitaare Zameen Par	280.00
Kiara Roy	War 2	400.00
Aditi Shah	HIT 3	310.00
Dev Yadav	Tourist Family	220.00
Riya Jain	Retro	240.00
Yash Saini	Thug Life	360.00
Mehak Khan	Lucifer 2	330.00
Aryan Mishra	Empuraan	350.00
Tanvi Agarwal	Marco	300.00
Kunal Malhotra	Deva	290.00
Pari Kaur	Bhool Chuk Maaf	240.00
Om Vyas	Kanguva	370.00
Diya Rao	Vidaamuyarchi	320.00
Rudra Saxena	Game Changer	340.00

20 rows in set (0.00 sec)

